Crowd Dynamics at Fringe Festival 2024: A Spatio–Temporal Model

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Background

“Festivals have increasingly become visitor events, and local people are crowded out by traffic, litter, queues, and overloaded infrastructure.”

-L Professor J John Lennon

"How might we effectively redistribute foot traffic in Edinburgh, ensuring popular landmarks remain accessible, enjoyable, and free from congestion-induced strain?"

Huge crowds throng the Royal Mile in August for free street entertainment during the Fringe. | other

Source:
1.https://www.hospitalityinvestor.com/strategy/cautious-pricing-critical-success-residences
2.https://www.edinburghnews.scotsman.com/whats-on/arts-and-entertainment/edinburghs-festivals-admit-action-needed-on-crowd-congestion-as-they-distance-themselves-from-growth-for-growths-sake-1832910

Motivation

Fringe festival is an important festival in Edinburgh.

How can we?...

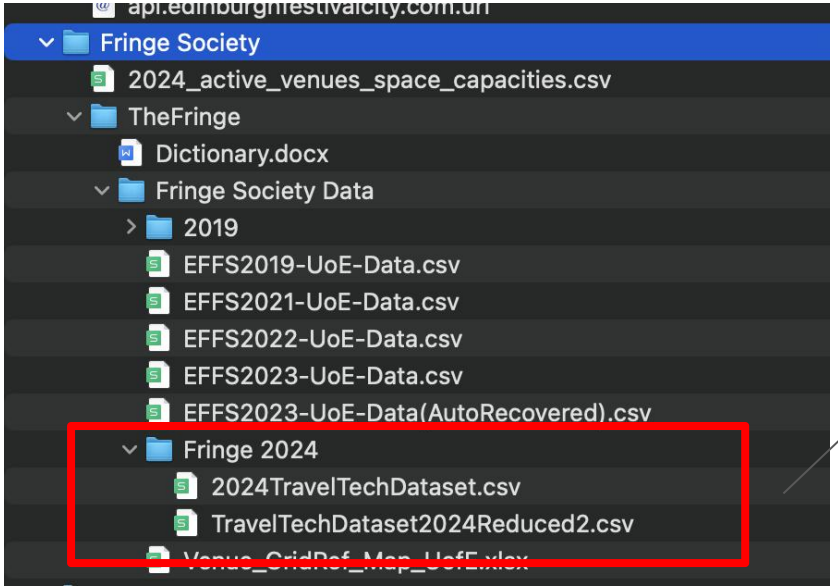
- …Understand pedestrian density (optimize site layout)
- …Reasonable allocation of facilities needs (toilets, garbage disposal, etc.)
- …Traffic flow (adjust routes or set up temporary stops, etc.)

Ways to visualize:

- ***Dynamic Insights:***
  Static visualizations fail to capture temporal changes in traffic flow, making it fool to reflect the change.
- ***Spatio-Temporal Complexity:***
  Understanding the interaction between space and time in traffic flow is challenging.
- ***Engaging Communication:***
  Use visual and physical representations to make complex data accessible and understandable for diverse audiences.

Dataset

Two datasets from the Fringe Society dataset is used in this project.



2024TravelTechDataset.csv

```
RangeIndex: 1856928 entries, 0 to 1856927
Data columns (total 17 columns):
```

all_2024_events_excel.xlsx

```
RangeIndex: 5155 entries, 0 to 5154
Data columns (total 32 columns):
```

2024TravelTechDataset.csv | all_2024_events_excel.xlsx

Delimiter: ,

	ticket_type	customer_id	customer_postcode
1	Concession	1544539	BT8 8EE
2	Concession	1544539	BT8 8EE
3	Concession	1544539	BT8 8EE

Methodology

Clean the dataset (2024)

- Modify to correct format

(e.g. the time column; change it to datetime format.)

Merge Datasets

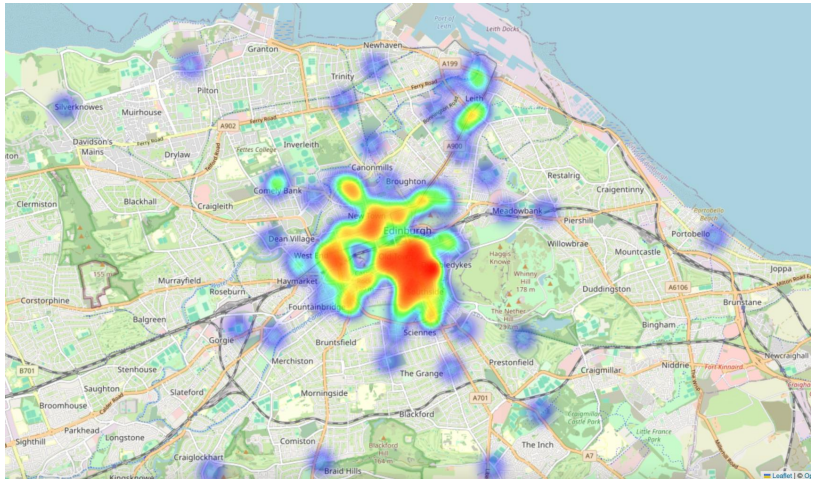
- Combine with the venue grid dataset using a left join.
- Remove duplicates/misleading entries.

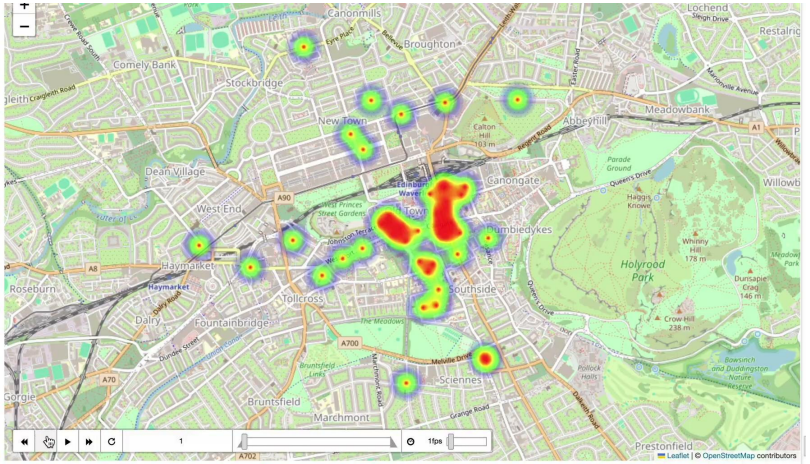
Generate Heatmap

- Use **Folium** to draw a heatmap.

Create Dynamic Visualization

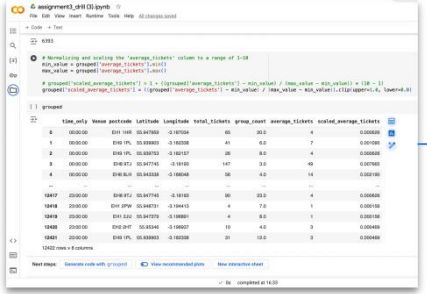
- Apply **HeatMapWithTime** to show daily changes.



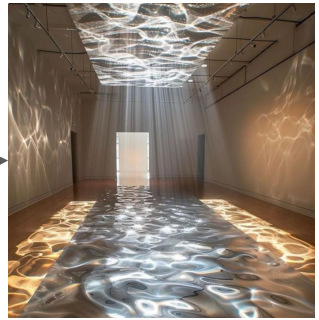


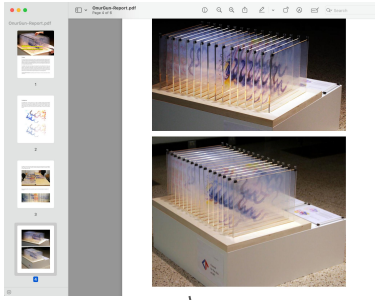
Workflow

1. Work on the dataset to create heatmaps

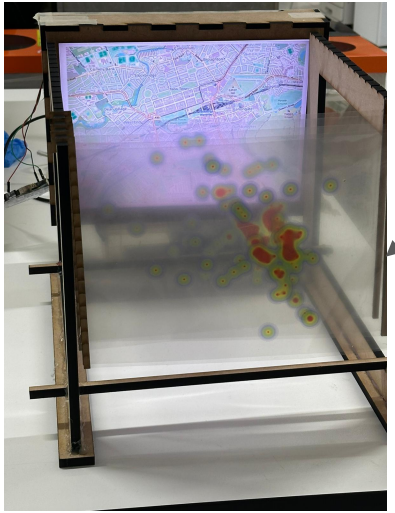


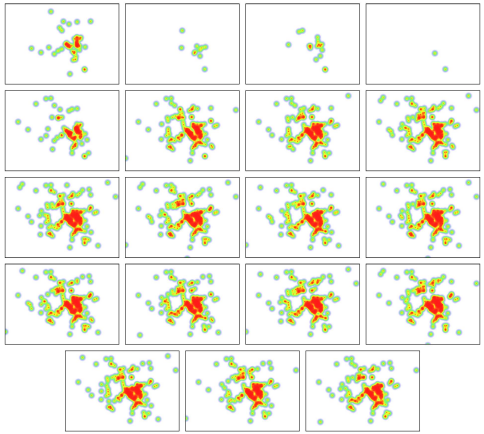
2. Explore ways to physicalize the dynamic data



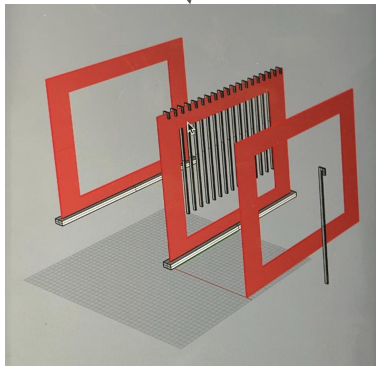


Gün, O. Y. (n.d.). *CAMIT Grant Program Report*. Design and Computation Group, Department of Architecture, MIT





4. Map heatmap images to transparent surfaces and assemble model



3. Design and build spatio-temporal model

Layering heatmaps to create a **Spatio-Temporal Model**

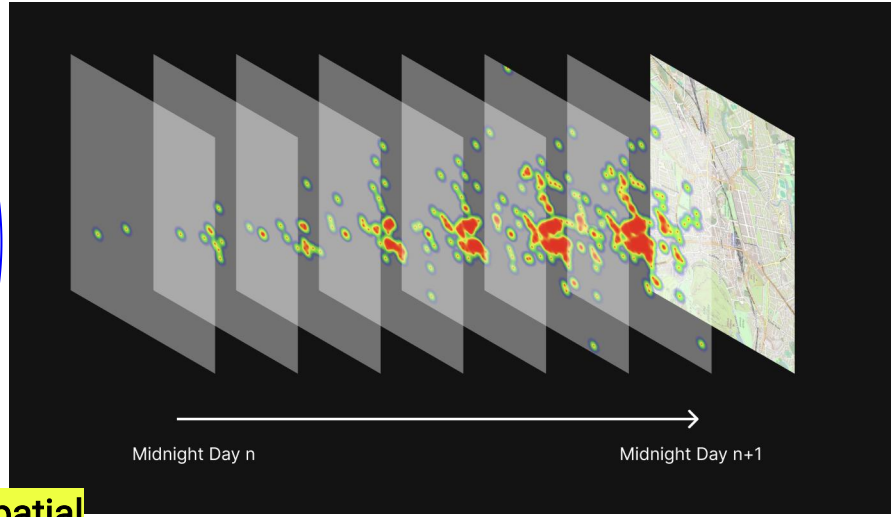
Spatio: Across space

(https://dictionary.cambridge.org/dictionary/english/spatio-temporal)

Data used: **longitude** and **latitude** of the Fringe events

Output: **pattern of crowd density** in each area of Edinburgh

>>>This creates a spatial dimension.



Temporal: Across time

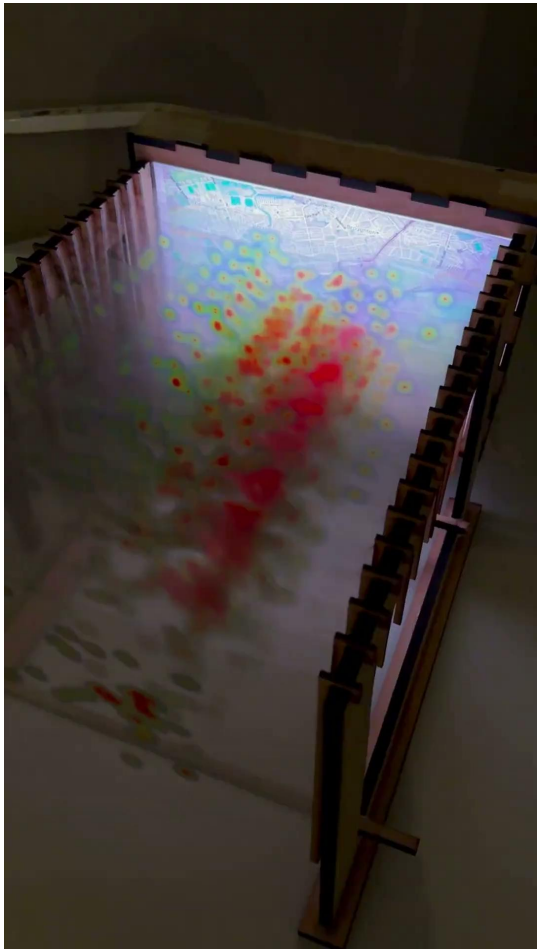
(https://dictionary.cambridge.org/dictionary/english/spatio-temporal)

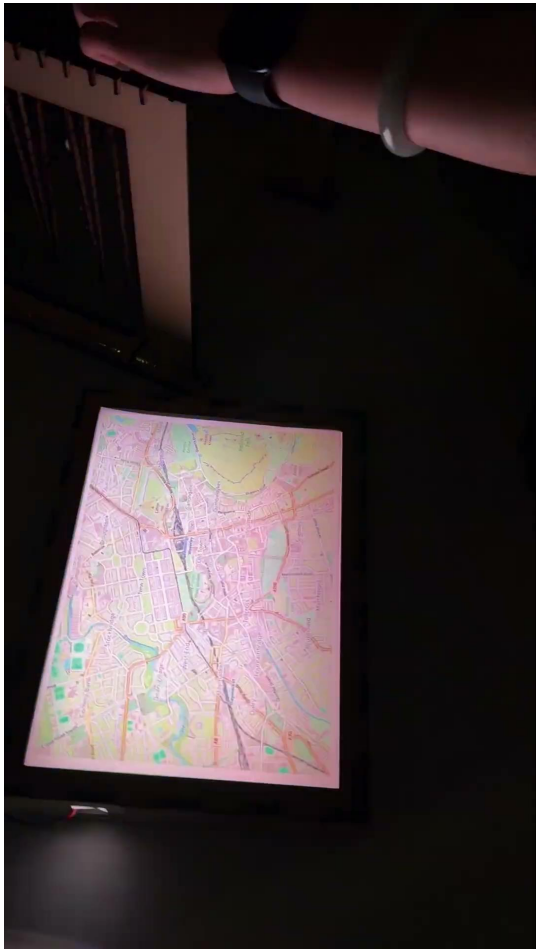
Data used: **average number of ticket purchases** for each Fringe event at **specific times** of the day throughout August 2024

Output: arranging them in sequential order, we visualized temporal movement.

>>>This creates a temporal dimension.

Each transparent layer that represents a heatmap of crowd density in Edinburgh Fringe Festival 2024 over the course of the day are arranged in sequential order. This way, the model becomes a spatio-temporal representation of how crowd density (spatial data) in different areas changes over time (temporal data), capturing patterns, relationships, and dynamics.





Results of physical model

Observers are able to experience the space-time of crowd movements throughout the day at the Fringe 2024.

Furthermore, each layer can be observed individually by placing it on top of the light box.

Results of data analysis

1. Identified hotspot trends for different times of day during the Fringe 2024 events
2. Highlighting popular zones/areas for the Fringe events

Summary

Problem Addressed: The project visualises and analyzes crowd density during the Fringe Festival 2024, providing insights into urban flow and congestion patterns.

Outcome: We combine a physical-digital hybrid to create a multi-sensory, interactive experience that engages users while communicating meaningful data-driven insights.

Significance:

- Helps tourists and locals- as well as festival organisers and urban planners- understand crowd trends.
- Encourages better management of congestion and exploration of less crowded areas.
- Highlights the potential of hybrid visualisations to represent complex dynamic datasets effectively.